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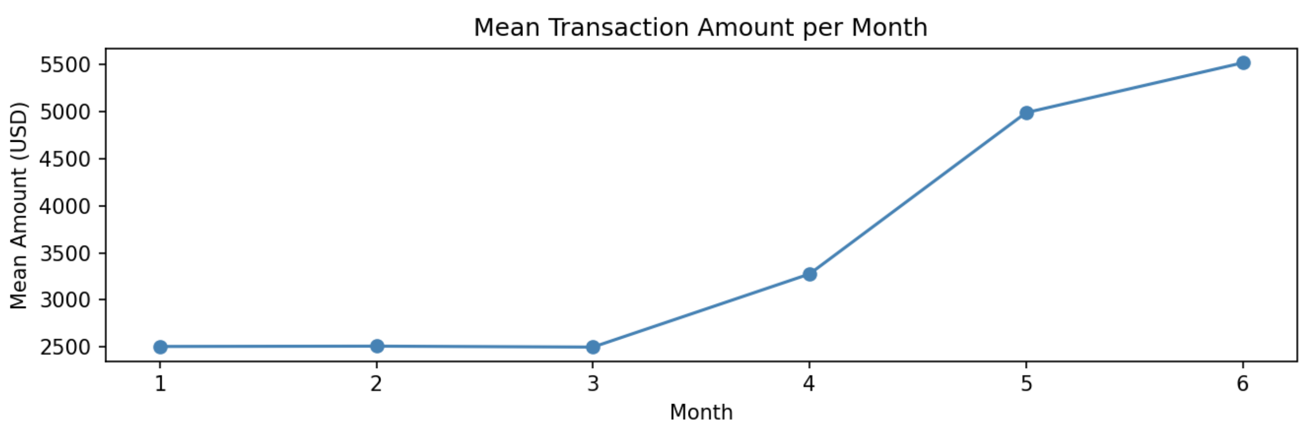
Course: Machine Learning   Section B

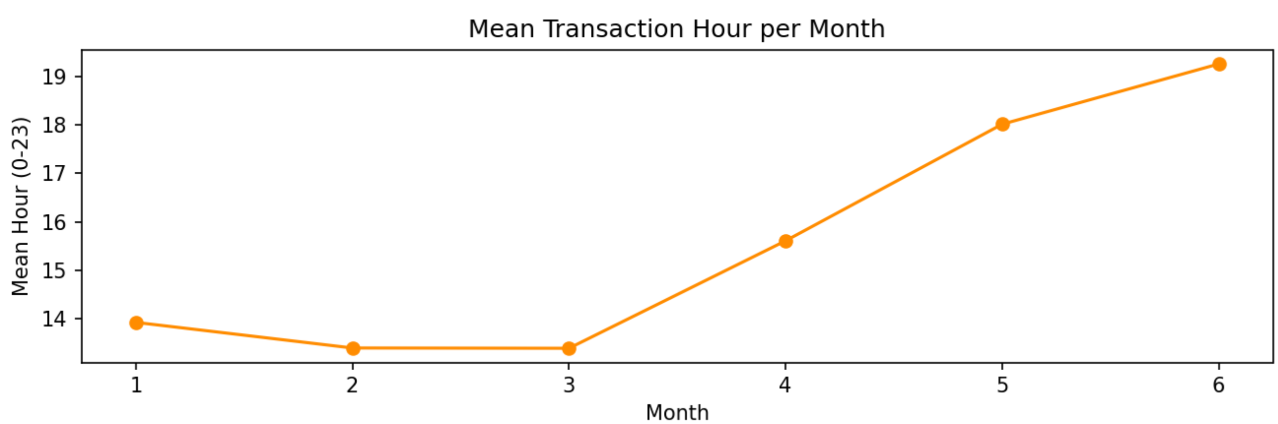
Task1:

```
● (.venv) ibrahim@MUFASA:~/Git/AI-ML/Lab13$ python lab13_solution.py

=== Task 1: Baseline Statistics ===
  transaction_amount: mean=2504.794, std=792.824, min=100.000, max=5582.190
  customer_age: mean=34.541, std=9.532, min=18.000, max=74.000
  transaction_hour: mean=13.570, std=4.065, min=2.000, max=23.000
  device_risk_score: mean=0.198, std=0.098, min=0.000, max=0.519
  Fraud rate: 0.0307
```

Task2:

```
=== Task 2: Monthly Batch Sizes ===
  Month 1: 500 rows | amount_mean=2505.86 | hour_mean=13.92
  Month 2: 500 rows | amount_mean=2509.23 | hour_mean=13.40
  Month 3: 500 rows | amount_mean=2499.30 | hour_mean=13.39
  Month 4: 500 rows | amount_mean=3275.25 | hour_mean=15.61
  Month 5: 500 rows | amount_mean=4987.20 | hour_mean=18.01
  Month 6: 500 rows | amount_mean=5516.07 | hour_mean=19.26
```





Task3:

```
=== Task 3: Mean-Shift Alerts ===
  Month 1: ok
  Month 2: ok
  Month 3: ok
  Month 4: ok
  Month 5: ALERT
  Month 6: ALERT
```

Task4:

```
=== Drift Event Log ===
 Month  Drift Phase               Batch Mean   Shift Mag   Sigmas Away
----------------------------------------------------------------------
     5  Feature Drift                4987.20     2482.40         3.131
     6  Concept Drift                5516.07     3011.28         3.798
```

Task5:

```
=== Task 5: KS Test Results ===
  Month 1: KS=0.0153, p=1.0000, drifted=False
  Month 2: KS=0.0193, p=0.9988, drifted=False
  Month 3: KS=0.0187, p=0.9993, drifted=False
  Month 4: KS=0.3640, p=0.0000, drifted=True
  Month 5: KS=0.8893, p=0.0000, drifted=True
  Month 6: KS=0.9333, p=0.0000, drifted=True
```

Task6:

```
=== Method Comparison ===
 Month   Mean-Shift Alert   KS Alert    KS Stat     p-value
------------------------------------------------------------
     1              False      False     0.0153      1.0000
     2              False      False     0.0193      0.9988
     3              False      False     0.0187      0.9993
     4              False       True     0.3640      0.0000
     5               True       True     0.8893      0.0000
     6               True       True     0.9333      0.0000
```

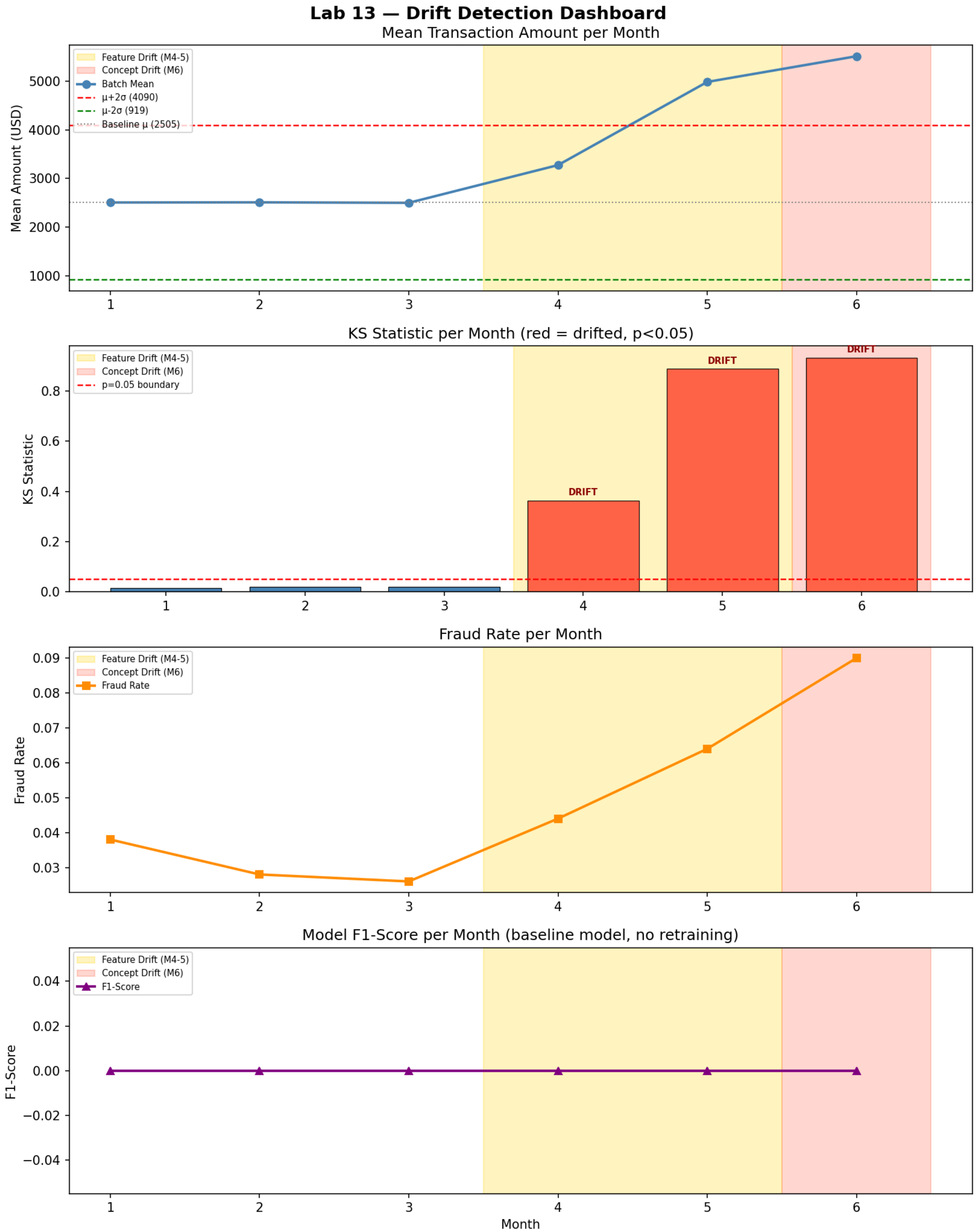
Task7:

```python
def task7_train_baseline_model(baseline_df):
    """
    Task 7: Train a Logistic Regression classifier on the baseline data.
    """
    X = baseline_df[FEATURES].values
    y = baseline_df[TARGET].values

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42)

    scaler  = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test  = scaler.transform(X_test)

    model = LogisticRegression(max_iter=1000, random_state=42)
    model.fit(X_train, y_train)

    y_pred   = model.predict(X_test)
    accuracy = float(accuracy_score(y_test, y_pred))
    f1       = float(f1_score(y_test, y_pred))

    return model, scaler, accuracy, f1
```

Task8:

```
=== Task 8: Model Performance per Month ===
  Month 1: Accuracy=0.9620, F1=0.0000
  Month 2: Accuracy=0.9720, F1=0.0000
  Month 3: Accuracy=0.9740, F1=0.0000
  Month 4: Accuracy=0.9560, F1=0.0000
  Month 5: Accuracy=0.9360, F1=0.0000
  Month 6: Accuracy=0.9100, F1=0.0000

--- Pipeline complete ---
Baseline model  Accuracy: 0.9700   F1: 0.0000
Mean-shift alerts : [5, 6]
KS-test alerts    : [4, 5, 6]
```

Task9:



Lab 13 — Drift Detection Dashboard
Mean Transaction Amount per Month
Feature Drift (M4-5)
Concept Drift (M6)
Batch Mean
μ+2σ (4090)
μ-2σ (919)
Baseline μ (2505)
Mean Amount (USD)
KS Statistic per Month (red = drifted, p<0.05)
p=0.05 boundary
KS Statistic
DRIFT
Fraud Rate per Month
Fraud Rate
Model F1-Score per Month (baseline model, no retraining)
F1-Score
Month

DISCUSSION ANSWERS

Q1. Feature drift vs. Concept drift (with dataset examples)?

Feature drift occurs when the INPUT distribution P(X) changes while the relationship P(Y|X) remains stable. In this dataset Months 4-5 show feature drift: transaction_amount and transaction_hour increased noticeably but the fraud detection rules that map these features to fraud labels did not fundamentally change, so a well-calibrated model may still perform reasonably.

Concept drift occurs when the relationship P(Y|X) itself changes, i.e. the same input patterns now map to different labels. Month 6 is the clearest example: the fraud rate tripled relative to baseline (new fraud patterns emerged), so the model's learned decision boundary is no longer valid even for similar feature values.

Q2. Month 4 type of drift and model impact?

Month 4 is purely feature drift (covariate shift). Transaction amounts rose but the fraud rate barely changed, meaning the learned mapping P(is_fraud | features) is still approximately correct. Logistic Regression generalises reasonably well to shifted inputs provided the class boundary has not moved, so accuracy and F1 should not drop dramatically in Month 4.

Q3. Why Month 6 causes a sharp F1 drop but not necessarily accuracy?

Accuracy is dominated by the majority class (legitimate transactions, ~97 % of baseline). If the model predicts "not fraud" for everything it still gets ~97 % accuracy. F1 specifically measures performance on the positive (fraud) class. When the fraud rate triples in Month 6, a new category of fraud patterns appears that the model was never trained on. The model misses many of these new frauds (high false negatives), causing precision and recall on the positive class and therefore F1 to collapse, even if overall accuracy stays deceptively high.

Q4. Why mean-shift misses drift that KS catches?

The mean-shift detector uses only the first moment (mean). If a distribution becomes bimodal or heavy-tailed while its mean stays close to baseline, the rule is blind to the shift. Example: suppose transaction amounts split into two groups many small amounts and a few very large ones such that the average is unchanged. The mean-shift detector fires no alert, yet the distributional shape has changed fundamentally. The KS test is sensitive to ANY difference in the full cumulative distribution, including changes in variance, skewness, or modality, so it catches such cases where the mean-shift rule fails.

Q5. Production ML response to detected drift?

Scheduled retraining with a sliding window: the system maintains a rolling window of recent labelled data (e.g. last 4-6 weeks). When the drift monitor fires (KS $p < 0.05$ or mean-shift alert), it triggers an automated retraining job that fits a fresh model on the most recent window, validates it on a held-out slice, and promotes it to production if it beats the incumbent on key metrics (accuracy, F1, AUC). This keeps the model aligned with the current data distribution without requiring manual intervention. Optionally, the system can weight recent samples more heavily (importance weighting) during retraining to further adapt to the new regime.